EMILIANO MORA

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EDUCATION

Illinois Institute of Technology, Chicago, IL Anticipated August 2026
Bachelor of Science - Mechanical Engineering
Pierce College, Puyallup, WA May 2024
Associate of Science – Engineering

SKILLS

- **CAD/Design:** SolidWorks, Inventor, REVIT, Revisto, NavisWorks, Altium Designer, GD&T, Technical Drawings
- **Programming/Controls:** Python, MATLAB, Simulink, Arduino C/C++, OpenCV
- **Robotics/Engineering:** Embedded Systems, Sensor Integration, Experimental Validation, FEA (ANSYS), LabVIEW

EXPERIENCE

Design Engineer
Gurtz Electric, Arlington Heights, IL October 2025 – April 2026
- Developed and revised 3D electrical layouts in Autodesk Revit for large-scale construction projects
- Interpreted multidisciplinary engineering drawings and resolved spatial conflicts through iterative design revisions
- Performed automated clash detection using NavisWorks and Revisto to identify integration risks and support downstream installation planning
- Optimized conduit routing strategies with consideration for prefabrication, installation constraints, and system accessibility

Operations Associate
United Performance Metals, Northbrook, IL May 2025 – October 2025
- Operated industrial plate saws producing custom cuts of nickel, titanium, stainless-steel, and aluminum stock within tolerances of $\pm 1/8$” to $\pm 1/16$”
- Supported manufacturing workflows involving high-performance engineering materials used in aerospace and industrial applications
- Revised SOPs to reduce setup time by 30% and improve single-operator workflow efficiency

Tutor
Pierce College, Puyallup, WA January 2023 - January 2024
- Hosted weekly group sessions in undergraduate engineering Statics, Dynamics, and Mechanics of Materials as well as tutoring appointments for Engineering Physics (1, 2, 3), and Chemistry Lab (1, 2, 3)

PROJECTS

Autonomous Flight and Validation of Crazyflie Microdrone
- Developed Python-based control and telemetry workflows for the Crazyflie 2.1+ microdrone using onboard state estimation and wireless communication interfaces
- Implemented and evaluated figure-8 trajectory tracking routines using Flowdeck v.2 localization and onboard sensing
- Created OpenCV-based post-processing and trajectory visualization tools to overlay logged flight data onto experimental flight footage for motion validation
- Investigated Model Predictive Control (MPC) frameworks including TinyMPC and acados for future embedded implementation on resource-constrained hardware
- Performed iterative debugging and hardware validation involving sensor drift, radio communication, and flight stability behavior

Bio-Inspired Walking Robot (Quadruped)
- Designed and fabricated a servo-actuated quadruped robot using laser-cut structural components and custom mechanical linkages
- Performed gait analysis and dynamic simulations to evaluate locomotion stability and linkage motion behavior
- Programmed Arduino-based embedded control logic in C/C++ to coordinate servo-driven leg motion
- Iterated mechanical layouts to improve manufacturability, assembly and structural performance

INTERESTS

Robotics, embedded systems, controls, electromechanical systems, rapid prototyping

REFERENCES

Available upon request